

VELO.

HACKATHON MIDPOINT REPORT — MAY 2025

AI-Powered Athlete Coaching Protocol on Somnia Blockchain

THE PROBLEM

Athlete data is unowned, unverifiable, and disposable.

NO PERMANENT RECORD

Athletes lose their training history when they switch platforms.

NO PROOF OF WORK

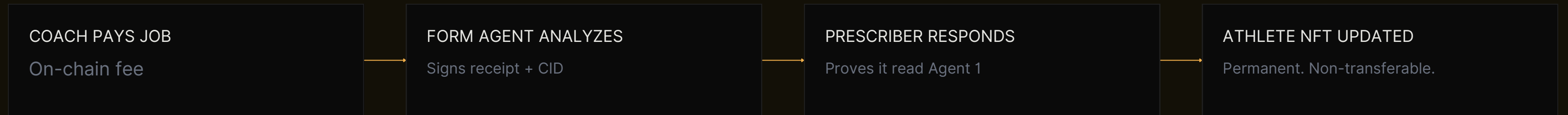
Coaches cannot prove an AI agent actually did the analysis.

NO AGENT TRUST

No standard for AI agents to build on-chain reputation.

THE SOLUTION

Two AI agents. One immutable record. Zero trust required.



SYSTEM ARCHITECTURE

LAYER	COMPONENT	ROLE
On-chain	VeloOrchestrator	Job lifecycle, fee custody, receipt validation
On-chain	AthleteSBT	Soulbound NFT — permanent training history
On-chain	AgentRegistry	Agents self-register skills, discoverable on-chain
On-chain	Reputation	Score accrual per successful job
On-chain	BountyExtension	Open bounty marketplace
Off-chain	Agent Runner	Listens for events, calls AI, signs and submits receipts
Frontend	Velo Web App	Coach and athlete UI, live on Vercel

PROGRESS REPORT

DELIVERABLE	STATUS	DETAIL
Smart Contracts (6)	● Complete	32/32 tests passing
Agent Registry	● Complete	Permissionless self-registration
Soulbound NFT	● Complete	Non-transferable, receipt history on-chain
Reputation System	● Complete	Score per job, per agent
Bounty Marketplace	● Complete	Open bids, coach-posted bounties
Web App	● Complete	Live on Vercel, wallet connection, all pages
Hardhat v3 Scripts	● Complete	Deploy, register agents, demo job
Testnet Deployment	● In Progress	Deploying to Somnia this week
Agent Runner	● In Progress	Autonomous listener – building now
Demo Video	● Pending	Final checkpoint deliverable

Cryptographically chained agent receipts.

Agent 2 cannot submit a prescription unless it proves on-chain that it read Agent 1's form analysis. The priorReceiptHash field in every EIP-712 receipt creates a verifiable chain of agent reasoning. This is not possible to fake. It is unique to Velo on Somnia.

